

Technology Stack

Date	15th July 2026
Team ID	SWTID-2026-379
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose

**1 Frontend /
Client-Side**

*HTML/CSS)
[Specify
technology]
[Explain why this was
chosen (e.g., responsive
design, component
reusability)]*

(e.g., React, Angular,

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python with Flask</i>	<i>Flask is a lightweight WSGI framework that integrates seamlessly with Python-based ML libraries (scikit-learn, XGBoost), making it easy to load the trained model and serve predictions through a simple API.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>Excel/CSV file-based storage (flood_dataset.xlsx) with Joblib/Pickle for model storage</i>	<i>Chosen because the project scope centers on model training and prediction rather than large-scale transactional data; the flat file format keeps data collection and preprocessing simple, and Pickle allows the trained model to be saved and reloaded efficiently.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>IBM Cloud</i>	<i>Selected for scalable, accessible deployment of the Flask web app, allowing disaster management teams to access flood predictions from any location without needing local installation.</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Used for source code management and team collaboration, allowing multiple team members (Afshan, Nehasini, Yagna, Sathwik) to work on different epics simultaneously and track changes through commits</i>

			<i>and pull requests.</i>
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6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Kaggle (dataset source), Anaconda Navigator, Jupyter Notebook</i>
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Kaggle provided the open-source rainfall/flood dataset used for training; Anaconda and Jupyter Notebook were used as the development environment for data analysis, model training, and testing before integrating into the Flask app.